engines. But muscles are like engines in that they often exert their force on cords, and levers with hinges. In us the levers are known as bones, the cords as tendons, and the hinges as joints. Quite a lot is known about the exact molecular ways in which muscles work, but I find more interesting the question of how muscle contractions are *timed*.

Have you ever watched an artificial machine of some complexity, a knitting or sewing machine, a loom, an automatic bottling factory, or a hay baler? Motive power comes from somewhere, an electric motor say, or a tractor. But much more baffling is the intricate timing of the operations. Valves open and shut in the right order, steel fingers deftly tie a knot round a hay bale, and then at just the right moment a knife shoots out and cuts the string. In many artificial machines timing is achieved by that brilliant invention the cam. This translates simple rotary motion into a complex rhythmic pattern of operations by means of an eccentric or specially shaped wheel. The principle of the musical box is similar. Other machines such as the steam organ and the pianola use paper rolls or cards with holes punched in a pattern. Recently there has been a trend towards replacing such simple mechanical timers with electronic ones. Digital computers are examples of large and versatile electronic devices which can be used for generating complex timed patterns of movements. The basic component of a modern electronic machine like a computer is the semiconductor, of which a familiar form is the transistor.

Survival machines seem to have bypassed the cam and the punched card altogether. The apparatus they use for timing their movements has more in common with an electronic computer, although it is strictly different in fundamental operation. The basic unit of biological computers, the nerve cell or neurone, is really nothing like a transistor in its internal workings. Certainly the code in which neurones communicate with each other seems to be a little bit like the pulse codes of digital computers, but the individual neurone is a much more sophisticated data-processing unit than the transistor. Instead of just three connections with other components, a single neurone may have tens of thousands. The neurone is slower than the transistor, but it has gone much further in the direction of miniaturization, a trend which has dominated the electronics industry over the past two decades. This is brought home by the fact that there are some ten thousand million neurones in the human brain: you could pack only a few hundred transistors into a skull.